

Hamiltonian-Geometric Optimization

A unifying geometric framework for training algorithms — validated, benchmarked, and honestly bounded

Sparsho Chakraborty Mohammad Alamgir Adreet Sarkar Nishanth M

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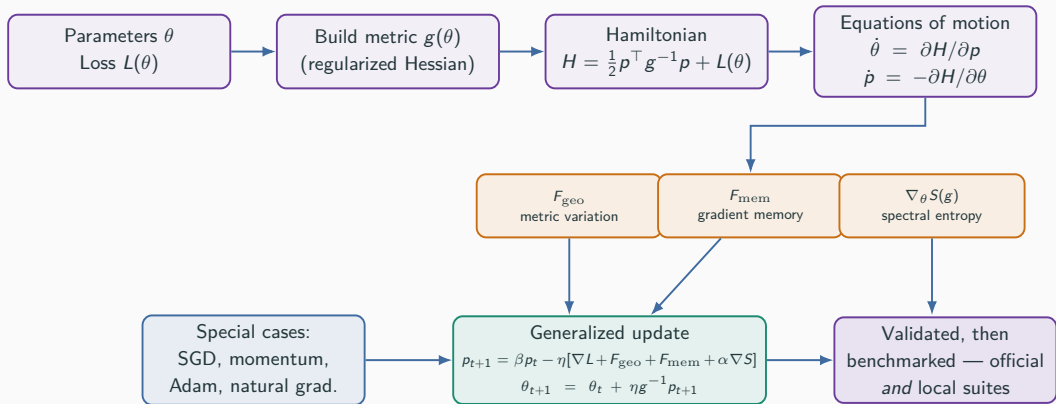
Hamiltonian Metric-Based Optimizer

Goal

Can a single **Hamiltonian-geometric** update rule act as a useful optimizer family across classical ML, LLM-style training, PINNs, and quantum-control objectives?

- Treat parameters θ as generalized coordinates, the loss $L(\theta)$ as potential energy, and optimizer state as momentum p evolving on a **metric manifold** $g(\theta)$.
- One family recovers SGD, momentum, Adam, and natural gradient as special cases.
- The claim is deliberately *bounded*: not uniformly superior, strongest when full-metric curvature is exploitable — and we test that boundary directly, not just assert it.

Methodology — Mathematical Framework at a Glance



Deterministic stages left to right; three correction forces merge into one update rule; output feeds equation-level validation before any benchmark is trusted.

The Update Rule — and Why Adam and SGD Are Special Cases of It

$$H(\theta, p) = \frac{1}{2} p^\top g^{-1}(\theta) p + L(\theta), \quad \dot{\theta} = \partial H / \partial p, \quad \dot{p} = -\partial H / \partial \theta$$
$$p_{t+1} = \beta p_t - \eta [\nabla_\theta L + F_{\text{geo}} + F_{\text{mem}} + \alpha \nabla_\theta S(g)], \quad \theta_{t+1} = \theta_t + \eta g^{-1} p_{t+1}$$

Optimizer	Hamiltonian-geometric specialization
SGD / heavy-ball	Flat metric, $F_{\text{geo}} = 0$
Adam / RMSProp	Diagonal adaptive metric; ignores metric-variation force
Natural gradient	Full information metric, no momentum/memory force
Hamiltonian-geometric	Full metric <i>with</i> momentum, memory, spectral entropy, curvature

Analytical Validation — Equation-Level, All 7 Checks PASS

Each row checks one equation from the design-reference literature review against a finite-difference computation on a seeded, genuinely curved-metric toy problem.

Check	Kind	Max error	Tol.	Result
Legendre transform	literal	2.96×10^{-11}	10^{-6}	PASS
Geometric force	literal	4.60×10^{-11}	10^{-5}	PASS
Flat-metric reduction	literal	0.000	10^{-9}	PASS
Memory recursion	literal	8.88×10^{-16}	10^{-9}	PASS
Hessian regularization	counterex.	0.000	10^{-9}	PASS
Rayleigh dissipation	completed	1.64×10^{-9}	10^{-6}	PASS
Spectral entropy gradient	completed	1.14×10^{-10}	10^{-5}	PASS

- Plus **Theorem 1**: on an exact-Hessian quadratic, the update is *provably* condition-number-independent for $0 < \eta < \sqrt{2}$, $0 \leq \beta < 1$ — a proof, not just a curve.

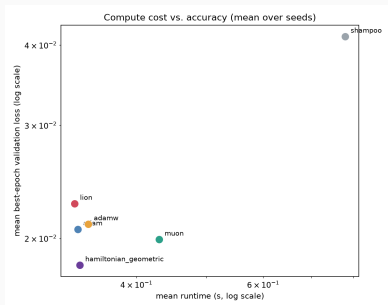
Headline Results — On *Official* Benchmark Codebases

Not reimplementations: the real deepobs package and the real MLCommons AlgoPerf codebase + dataset, run end-to-end with our optimizer wired into their own APIs.

	Optimizer	Val. acc.	Test acc.
Official DeepOBS (mnist_mlp, 20 epochs, tuned LR each)	Hamiltonian-geometric	0.985	0.985
	SGD-momentum	0.982	0.984
	AdamW	0.978	0.980
	Optimizer	Val. acc.	Test acc.
Official AlgoPerf (OGBG workload, 20 steps)	AdamW	0.984	0.983
	Hamiltonian-geometric	0.983	0.983
	SGD	0.955	0.949

- **Best of three on real official DeepOBS; a genuine near-tie with AdamW, both clearly ahead of SGD, on real official AlgoPerf.**
- AlgoPerf run is reduced in scale (20 of $\sim 11,300$ official steps; smaller eval batch — 4GB GPU, not the official $4\times A100$ rig) and disclosed as such, not hidden.

Broader Evidence — Classical, Modern-Optimizer, and Honest About Seeds

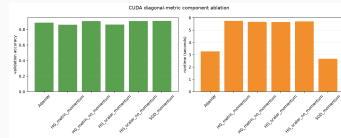
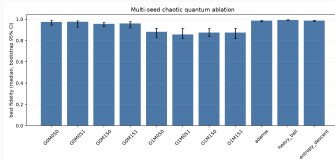
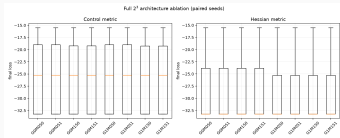


- Classical suite (MNIST, AlgoPerf-/DeepOBS-style local, PINN): best or near-best across the board.
- Modern-optimizer benchmark (spiral MLP): Adam, HG, and Muon are within 4×10^{-4} validation loss — a leading group, not a decisive win.

We checked ourselves, and it wasn't all favorable

A 10/10/10-seed replication shows one earlier local-benchmark win (DeepOBS-style) does *not* replicate ($p = 0.69$, a coin flip) — and a small-LLM loss to AdaFactor is real and significant ($p = 0.003$), not the near-tie first reported.

Extended Ablation — Where the Extra Machinery Helps, and Where It Doesn't



- **Architecture (40 seeds):** the spectral term is significant only under the *genuine* Hessian metric ($p = 9.25 \times 10^{-5}$), inert under an unrelated one — the mechanism is real, not decorative.
- **Quantum control & GPU scaling (10 seeds each):** the honest negative — simple baselines beat every HG variant here, at a fraction of the runtime. Reported, not hidden.

Takeaways

- Formulation is mathematically coherent and now partly **proven** (Theorem 1), not just simulated.
- Best-of-three on **real official DeepOBS**; a genuine near-tie with AdamW on **real official AlgoPerf** — the two cleanest external benchmarks available to us.
- Strongest exactly where theory predicts: genuine curvature. Weakest exactly where theory predicts: quantum control, GPU-scale diagonal reduction.
- Every favorable result here has a matching stress test that could have — and in two cases did — go the other way.

Thank You

Hamiltonian-Geometric Optimization Framework